

16.07 Dynamics

Problem Set 2 — Solutions

Due Monday, October 5

Notation. Throughout, $s \equiv \sqrt{\ell^2 - x^2 - y^2}$ is the depth of the mass below the pivot, $\omega_0 \equiv \sqrt{g/\ell}$ is the pendulum frequency, and

$$\Omega_n \equiv \Omega \cos \lambda, \quad \Omega_u \equiv \Omega \sin \lambda$$

are the north and up components of the Earth's angular velocity. We will also use the 2×2 skew-symmetric matrix

$$\mathbf{S} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{S}^T = -\mathbf{S}, \quad \mathbf{S}^2 = -\mathbf{I}.$$

Multiplying a vector by $\mathbf{S}$ rotates it 90° counterclockwise. Section 2 explains how $\mathbf{S}$ relates to the skew-symmetric matrix $\mathbf{S}(\vec{\omega})$ used for 3D rotations. With $\ell = 10$ m and $g = 9.81$ m/s², $\omega_0 = 0.9905$ rad/s, and the swing period is $2\pi/\omega_0 = 6.34$ s.

1 Deriving Equations of Motion

(a) Energies

The Earth's rotation axis lies in the plane spanned by $\hat{n}$ and $\hat{u}$ and makes angle λ with the local horizontal (Fig. 1, left). So

$$\vec{\Omega} = \Omega \cos \lambda \hat{n} + \Omega \sin \lambda \hat{u} = \begin{bmatrix} 0 \\ \Omega_n \\ \Omega_u \end{bmatrix}_{\text{ENU}}.$$

(Check: at the North Pole, $\lambda = 90^\circ$ and $\vec{\Omega}$ points straight up; at the equator, it points due north.)

Ignoring the translation of P , the inertial velocity of the mass is the velocity seen in the rotating frame plus the transport term:

$$\vec{v} = \dot{\vec{r}} + \vec{\Omega} \times \vec{r}, \quad \vec{\Omega} \times \vec{r} = \begin{vmatrix} \hat{e} & \hat{n} & \hat{u} \\ 0 & \Omega_n & \Omega_u \\ x & y & z \end{vmatrix} = (\Omega_n z - \Omega_u y) \hat{e} + \Omega_u x \hat{n} - \Omega_n x \hat{u}.$$

The kinetic energy is

$$T = \frac{1}{2} m |\vec{v}|^2 = \frac{1}{2} m [(\dot{x} + \Omega_n z - \Omega_u y)^2 + (\dot{y} + \Omega_u x)^2 + (\dot{z} - \Omega_n x)^2].$$

Expanding and dropping the $\mathcal{O}(\Omega^2)$ terms,

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + m \Omega_u (x \dot{y} - y \dot{x}) + m \Omega_n (z \dot{x} - x \dot{z}), \quad U = mgz.$$

Why the $\mathcal{O}(\Omega^2)$ terms are negligible. The dropped terms are centrifugal, of size $\sim \frac{1}{2}m\Omega^2\ell^2$, while the gravitational terms are $\sim mg\ell$. The ratio is $\Omega^2\ell/g \approx 5 \times 10^{-9}$. By contrast, the $\mathcal{O}(\Omega)$ (Coriolis) terms are down by only $\Omega/\omega_0 \approx 7 \times 10^{-5}$, and they act over tens of thousands of swings. They are what produce the precession.

(b) Eliminating the constraint

Since the mass hangs below the pivot, $z < 0$, so we take the negative root:

$$z = -\sqrt{\ell^2 - x^2 - y^2} = -s, \quad \dot{z} = \frac{x\dot{x} + y\dot{y}}{s}.$$

Substituting, the Ω_n term becomes

$$z\dot{x} - x\dot{z} = -s\dot{x} - \frac{x(x\dot{x} + y\dot{y})}{s} = -\frac{(\ell^2 - y^2)\dot{x} + xy\dot{y}}{s},$$

using $s^2 + x^2 = \ell^2 - y^2$. The Lagrangian $L = T - U$ is

$$L = \frac{1}{2}m \left[\dot{x}^2 + \dot{y}^2 + \frac{(x\dot{x} + y\dot{y})^2}{s^2} \right] + m\Omega_u(x\dot{y} - y\dot{x}) - m\Omega_n \frac{(\ell^2 - y^2)\dot{x} + xy\dot{y}}{s} + mgs.$$

(c) Linearization via the Lagrangian

Expand each term to second order in $(x, y, \dot{x}, \dot{y})$, using

$$s = \ell - \frac{x^2 + y^2}{2\ell} + \mathcal{O}(4), \quad \frac{1}{s} = \frac{1}{\ell} + \mathcal{O}(2).$$

- $\frac{1}{2}m(\dot{x}^2 + \dot{y}^2)$ is already quadratic. The term $(x\dot{x} + y\dot{y})^2/s^2$ is fourth order, so we drop it.
- $m\Omega_u(x\dot{y} - y\dot{x})$ is already quadratic.
- $-m\Omega_n [(\ell^2 - y^2)\dot{x} + xy\dot{y}]/s = -m\Omega_n\ell\dot{x} + \mathcal{O}(3)$.
- $mgs = mg\ell - \frac{mg}{2\ell}(x^2 + y^2) + \mathcal{O}(4)$.

The quadratic Lagrangian is therefore

$$L_2 = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) + m\Omega_u(x\dot{y} - y\dot{x}) - \frac{mg}{2\ell}(x^2 + y^2) - m\Omega_n\ell\dot{x} + \text{const.}$$

The term $-m\Omega_n\ell\dot{x}$ is first order, but it will drop out in the Euler-Lagrange equation when we compute $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right)$, so we drop it.

Equations of motion.

$$\begin{aligned} \frac{\partial L_2}{\partial \dot{x}} = m\dot{x} - m\Omega_u y, \quad \frac{\partial L_2}{\partial x} = m\Omega_u \dot{y} - \frac{mg}{\ell} x &\implies m\ddot{x} - 2m\Omega_u \dot{y} + \frac{mg}{\ell} x = 0, \\ \frac{\partial L_2}{\partial \dot{y}} = m\dot{y} + m\Omega_u x, \quad \frac{\partial L_2}{\partial y} = -m\Omega_u \dot{x} - \frac{mg}{\ell} y &\implies m\ddot{y} + 2m\Omega_u \dot{x} + \frac{mg}{\ell} y = 0. \end{aligned}$$

In the form $\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0}$:

$$\mathbf{M} = m\mathbf{I}, \quad \mathbf{C} = 2m\Omega \sin \lambda \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = 2m\Omega_u \mathbf{S}, \quad \mathbf{K} = \frac{mg}{\ell} \mathbf{I}.$$

Structure of $\mathbf{C}$. $\mathbf{C}$ is *skew-symmetric*: $\mathbf{C}^T = -\mathbf{C}$. Consequences:

- $\dot{\mathbf{q}}^T \mathbf{C} \dot{\mathbf{q}} = 0$, so the force $-\mathbf{C} \dot{\mathbf{q}}$ is always perpendicular to the velocity. It does no work and causes no damping.
- Only the vertical component $\Omega_u = \Omega \sin \lambda$ appears. For horizontal velocities, the horizontal component $\Omega_n \hat{n}$ produces a Coriolis force $-2m\Omega_n \hat{n} \times (\dot{x}\hat{e} + \dot{y}\hat{n}) = 2m\Omega_n \dot{x} \hat{u}$, which is vertical. That force is absorbed by the cable tension.

2 Analyzing Precession

Dividing Eq. (1) by m gives

$$\ddot{\mathbf{q}} + 2\Omega_u \mathbf{S} \dot{\mathbf{q}} + \omega_0^2 \mathbf{q} = \mathbf{0}.$$

Take $\mathbf{R}(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{bmatrix}$, which rotates counterclockwise by ψ , consistent with the definition of ψ . Let the frame rotate at a constant rate, $\psi(t) = \dot{\psi} t$.

Derivative of a rotation matrix. Recall that in 3D, a rotation matrix $\mathbf{R}(t)$ whose frame rotates with angular velocity $\vec{\omega}$ (expressed in the rotating frame) satisfies

$$\dot{\mathbf{R}} = \mathbf{R} \mathbf{S}(\vec{\omega}), \quad \mathbf{S}(\vec{\omega}) = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix},$$

where $\mathbf{S}(\vec{\omega})$ is the skew-symmetric matrix with $\mathbf{S}(\vec{\omega}) \mathbf{v} = \vec{\omega} \times \mathbf{v}$. The 2D case is the special case of rotation about a fixed axis perpendicular to the plane, $\vec{\omega} = [0 \ 0 \ \omega]^T$. Keeping only the in-plane 2×2 block, $\mathbf{S}(\omega)$ depends on the single scalar ω and factors as

$$\mathbf{S}(\omega) = \begin{bmatrix} 0 & -\omega \\ \omega & 0 \end{bmatrix} = \mathbf{S} \omega, \quad \text{so} \quad \dot{\mathbf{R}} = \mathbf{R} \mathbf{S}(\omega) = \mathbf{R} \mathbf{S} \omega.$$

Here $\omega = \dot{\psi}$, so $\dot{\mathbf{R}} = \mathbf{R} \mathbf{S} \dot{\psi}$. Unlike in 3D, every 2D rotation shares the same axis, so $\mathbf{S}$ commutes with $\mathbf{R}$: $\mathbf{S} \mathbf{R} = \mathbf{R} \mathbf{S}$. (You can also verify this directly by multiplying out the matrices.) These two facts make the algebra short. Differentiating $\mathbf{q} = \mathbf{R} \tilde{\mathbf{q}}$ twice:

$$\begin{aligned} \dot{\mathbf{q}} &= \dot{\mathbf{R}} \tilde{\mathbf{q}} + \mathbf{R} \dot{\tilde{\mathbf{q}}} = \mathbf{R} (\dot{\tilde{\mathbf{q}}} + \mathbf{S} \tilde{\mathbf{q}} \dot{\psi}), \\ \ddot{\mathbf{q}} &= \mathbf{R} \mathbf{S} \dot{\psi} (\dot{\tilde{\mathbf{q}}} + \mathbf{S} \tilde{\mathbf{q}} \dot{\psi}) + \mathbf{R} (\ddot{\tilde{\mathbf{q}}} + \mathbf{S} \dot{\tilde{\mathbf{q}}} \dot{\psi} - \tilde{\mathbf{q}} \dot{\psi}^2) = \mathbf{R} (\ddot{\tilde{\mathbf{q}}} + 2\mathbf{S} \dot{\tilde{\mathbf{q}}} \dot{\psi} - \tilde{\mathbf{q}} \dot{\psi}^2), \end{aligned}$$

where we used $\ddot{\psi} = 0$ and $\mathbf{S}^2 = -\mathbf{I}$. Substituting into the equation of motion:

$$\mathbf{R} [\ddot{\tilde{\mathbf{q}}} + 2\mathbf{S} \dot{\tilde{\mathbf{q}}} \dot{\psi} - \tilde{\mathbf{q}} \dot{\psi}^2] + 2\Omega_u \mathbf{S} \mathbf{R} [\dot{\tilde{\mathbf{q}}} + \mathbf{S} \tilde{\mathbf{q}} \dot{\psi}] + \omega_0^2 \mathbf{R} \tilde{\mathbf{q}} = \mathbf{0}.$$

Moving $\mathbf{S}$ past $\mathbf{R}$, using $\mathbf{S}^2 = -\mathbf{I}$, and multiplying on the left by $\mathbf{R}^T$:

$$\boxed{\ddot{\tilde{\mathbf{q}}} + 2(\dot{\psi} + \Omega_u) \mathbf{S} \dot{\tilde{\mathbf{q}}} + (\omega_0^2 - \dot{\psi}^2 - 2\Omega_u \dot{\psi}) \tilde{\mathbf{q}} = \mathbf{0}.}$$

The rotating frame contributes its own velocity-rotation term, $2\mathbf{S} \dot{\tilde{\mathbf{q}}} \dot{\psi}$, with the same structure as the Coriolis term. These two terms can cancel. (This step relies on $\mathbf{M}$ and $\mathbf{K}$ being multiples of $\mathbf{I}$, so they commute with $\mathbf{R}$. The linearized pendulum has no preferred horizontal direction.)

(a) Precession rate

The $\mathbf{S}\dot{\mathbf{q}}$ term vanishes when

$$\dot{\psi} = -\Omega \sin \lambda.$$

With this choice, the remaining ODE is

$$\ddot{\tilde{\mathbf{q}}} + (\omega_0^2 + \Omega^2 \sin^2 \lambda) \tilde{\mathbf{q}} = \mathbf{0} \approx \ddot{\tilde{\mathbf{q}}} + \omega_0^2 \tilde{\mathbf{q}} = \mathbf{0},$$

since $\Omega^2 \sin^2 \lambda / \omega_0^2 \lesssim 5 \times 10^{-9}$. (This is consistent with having dropped $\mathcal{O}(\Omega^2)$ terms in 1(a).) In the rotating frame, the mass behaves as an ordinary 2D pendulum. Its solutions $\tilde{\mathbf{q}}(t)$ trace a fixed line or ellipse that does not rotate. Back in the Earth frame, $\mathbf{q} = \mathbf{R}(-\Omega_u t) \tilde{\mathbf{q}}$ is that same fixed pattern turned steadily at rate $-\Omega_u$. So the swing plane precesses at $\dot{\psi} = -\Omega \sin \lambda$.

(b) Direction

- **Northern hemisphere** ($\lambda > 0$): $\dot{\psi} < 0$. Since ψ is measured counterclockwise from above, the swing plane rotates *clockwise* viewed from above. It rotates counterclockwise in the southern hemisphere.
- **Equator** ($\lambda = 0$): $\dot{\psi} = 0$, so there is no precession. $\vec{\Omega}$ is horizontal there, and horizontal rotation only produces vertical Coriolis forces on the (nearly horizontal) swing.
- **North Pole** ($\lambda = 90^\circ$): $\dot{\psi} = -\Omega$, one full clockwise turn per sidereal day. At the pole, the pivot is on the rotation axis and gravity is parallel to it, so nothing can exert a torque that rotates the swing plane in inertial space. The plane stays fixed relative to the stars while the Earth turns counterclockwise beneath it.

(c) Precession period at MIT

With $\lambda = 42.36^\circ$, $\sin \lambda = 0.6737$:

$$|\dot{\psi}| = \Omega \sin \lambda = 4.913 \times 10^{-5} \text{ rad/s} = 10.13^\circ/\text{h}, \quad T_{\text{prec}} = \frac{2\pi}{\Omega \sin \lambda} = 1.279 \times 10^5 \text{ s} \approx \boxed{35.5 \text{ h.}}$$

3 Simulation

The code is listed at the end of this document. It integrates the linearized equations from 1(c) for 36 hours (just over one precession period, about 20,000 swings) using `solve_ivp` with the `DOP853` method and relative and absolute tolerances of 10^{-10} . The mass starts at rest with $x(0) = \ell \sin 5^\circ = 0.872 \text{ m}$ and $y(0) = 0$. (Because the equations are linear, starting from $\ell\theta_0$ instead gives identical precession.)

Energy check. Because $\mathbf{C}$ is skew-symmetric, the Earth-frame energy

$$E = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}} + \frac{1}{2} \mathbf{q}^T \mathbf{K} \mathbf{q}$$

is conserved exactly: $\dot{E} = \dot{\mathbf{q}}^T (\mathbf{M} \ddot{\mathbf{q}} + \mathbf{K} \mathbf{q}) = -\dot{\mathbf{q}}^T \mathbf{C} \dot{\mathbf{q}} = 0$. Over 36 hours, the maximum relative drift in E was 8×10^{-7} , confirming the integration is accurate.

Computing ψ . At each turning point (local maximum of $r = \sqrt{x^2 + y^2}$), the orientation of the swing line is computed as

$$\psi = \frac{1}{2} \text{atan2}(2xy, x^2 - y^2),$$

which gives the angle of the line through the origin and (x, y) modulo π . It does not care which end of the swing the mass is on. The π jumps are then removed with $\psi_{\text{unwrapped}} = \frac{1}{2} \text{unwrap}(2\psi)$. An equivalent approach is to use $\text{atan2}(y, x)$ and unwrap with a period of π .

Results. Figure 1 shows the simulated $\psi(t)$ lying on top of the prediction. A linear fit gives

$$\dot{\psi}_{\text{sim}} = -10.13445^\circ/\text{h} \quad \text{vs.} \quad \dot{\psi}_{\text{pred}} = -10.13445^\circ/\text{h}, \quad \text{error} \approx 6 \times 10^{-8} \, \%.$$

The agreement is at the level of numerical error.

Grading note: Errors much larger than about $10^{-3}\%$ usually indicate loose integration tolerances, too coarse an output time step (so the turning points are poorly located), or angle-wrapping mistakes. Visible jumps of $\pm 180^\circ$ or $\pm 90^\circ$ in $\psi(t)$ mean the unwrapping was not done with period π .

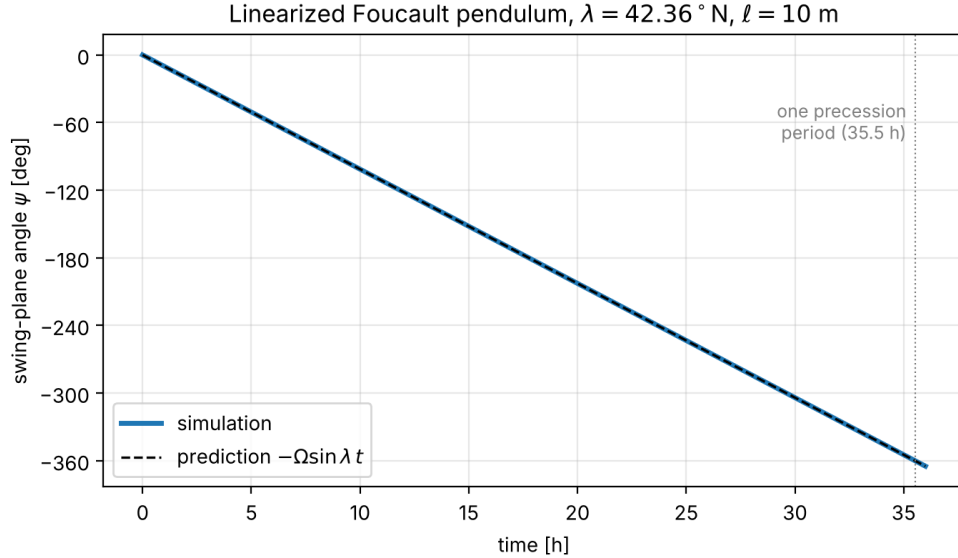


Figure 1: Simulated swing-plane angle ψ (evaluated once per swing at the turning points) and the prediction $\psi = -\Omega \sin \lambda t$ for $\lambda = 42.36^\circ \text{N}$. The dotted line marks one precession period, 35.5 h.

Code

```

1 """16.07 Problem Set 2 -- Foucault pendulum simulation (solution code).
2
3 Integrates the linearized equations of motion from Section 1(c),
4      $M \ddot{q} + C \dot{q} + K q = 0$ ,  $q = [x, y]$  (east, north),
5 with  $M = m I$ ,  $C = 2 m \Omega \sin(\text{lat}) J$ ,  $K = (m g / \ell) I$ , and compares the
6 precession of the swing plane with the prediction  $\psi_{\text{dot}} = -\Omega \sin(\text{lat})$ .
7 """
8 import numpy as np

```

```

9 import matplotlib.pyplot as plt
10 from scipy.integrate import solve_ivp
11
12 # ----- Parameters -----
13 m = 1.0 # kg (cancels out of the dynamics)
14 g = 9.81 # m/s^2
15 ell = 10.0 # m
16 Omega = 7.2921e-5 # rad/s (sidereal)
17 lat = np.deg2rad(42.36) # MIT
18 Ou = Omega * np.sin(lat) # vertical component of Earth's rotation
19 w0sq = g / ell
20
21 psi_dot_pred = -Ou
22 T_prec = 2 * np.pi / Ou # time for the swing plane to turn 360 deg
23 T_final = 36 * 3600.0 # > one full precession period (35.5 h)
24
25 J = np.array([[0.0, -1.0], [1.0, 0.0]])
26 M = m * np.eye(2)
27 C = 2 * m * Ou * J
28 K = m * w0sq * np.eye(2)
29 Minv = np.linalg.inv(M)
30
31
32 def dynamics(t, X):
33     q, qd = X[:2], X[2:]
34     qdd = -Minv @ (C @ qd + K @ q)
35     return np.concatenate([qd, qdd])
36
37
38 def energy(X):
39     """Earth-frame energy  $E = 1/2 \dot{q}^T M \dot{q} + 1/2 q^T K q$  (conserved since  $C = -C^T$ )."""
40     q, qd = X[:2], X[2:]
41     return 0.5 * np.einsum("i...,ij,j...->...", qd, M, qd) + \
42         0.5 * np.einsum("i...,ij,j...->...", q, K, q)
43
44
45 def swing_angle(t, x, y):
46     """Swing-plane angle  $\psi$  at the turning points (local maxima of  $r$ )."""
47     r = np.hypot(x, y)
48     k = np.where((r[1:-1] > r[:-2]) & (r[1:-1] >= r[2:]))[0] + 1
49     psi = 0.5 * np.arctan2(2 * x[k] * y[k], x[k]**2 - y[k]**2) # angle of the line, mod pi
50     psi = 0.5 * np.unwrap(2 * psi) # remove the pi jumps
51     return t[k], psi
52
53
54 if __name__ == "__main__":
55     theta0 = np.deg2rad(5.0)
56     X0 = [ell * np.sin(theta0), 0.0, 0.0, 0.0] # released from rest, displaced east
57     t_eval = np.arange(0.0, T_final, 0.05)
58     sol = solve_ivp(dynamics, [0, T_final], X0, method="DOP853",
59                     t_eval=t_eval, rtol=1e-10, atol=1e-10)
60
61     # Energy check
62     E = energy(sol.y)
63     print(f"max relative energy drift: {np.max(np.abs(E - E[0])) / E[0]:.2e}")
64
65     # Precession rate
66     tk, psi = swing_angle(sol.t, sol.y[0], sol.y[1])
67     rate = np.polyfit(tk, psi, 1)[0]

```

```

68 err = 100 * (rate - psi_dot_pred) / psi_dot_pred
69 print(f"predicted: {np.rad2deg(psi_dot_pred) * 3600:.6f} deg/h, "
70       f"period {T_prec / 3600:.3f} h")
71 print(f"measured: {np.rad2deg(rate) * 3600:.6f} deg/h, error {err:.2e} %")
72
73 # Plot
74 fig, ax = plt.subplots(figsize=(7, 4.2))
75 ax.plot(tk / 3600, np.rad2deg(psi), "C0", lw=2.5, label="simulation")
76 ax.plot(tk / 3600, np.rad2deg(psi_dot_pred * tk), "k--", lw=1.2,
77       label=r"prediction  $-\Omega \sin \lambda, t$ ")
78 ax.axvline(T_prec / 3600, color="gray", lw=0.8, ls=":")
79 ax.text(T_prec / 3600 - 0.4, -40, "one precession\nperiod (35.5 h)",
80       ha="right", va="top", fontsize=9, color="gray")
81 ax.set_xlabel("time [h]")
82 ax.set_ylabel(r"swing-plane angle  $\psi$  [deg]")
83 ax.set_yticks(np.arange(0, -361, -60))
84 ax.set_title(r"Linearized Foucault pendulum,  $\lambda = 42.36^\circ$ ,  $\ell = 10$  m")
85 ax.grid(alpha=0.3)
86 ax.legend(loc="lower left")
87 fig.tight_layout()
88 fig.savefig("psi_linear.png", dpi=200)

```